

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 04

DATE : 09-06-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 04

Topics

DATE : 09 - 06 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]

PHYSICS : Electrostatics, Gravitation, Verticle Cercular Motion, WEP

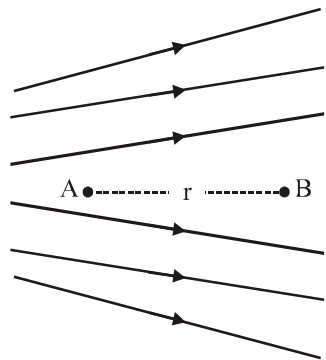
CHEMISTRY : Organic Chemistry, Equilibrium, s-Block

BIOLOGY : Genetics-I, Protista and Fungi, Complete Human Reproduction, Nervous System

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

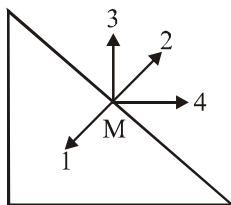
- Two small conducting spheres of equal radius have charges $+10\mu\text{C}$ and $-20\mu\text{C}$ respectively and placed at a distance R from each other experience force F_1 . If they are brought in contact and separated to the same distance, they experience force F_2 . The ratio of F_1 to F_2 is:-
 (1) $1 : 8$ (2) $-8 : 1$
 (3) $1 : 2$ (4) $-2 : 1$
- Equal charges q are placed at the four corners A, B, C, D of a square of length a . The magnitude of the force on the charge at B will be:-
 (1) $\frac{3q^2}{4\pi\epsilon_0 a^2}$
 (2) $\frac{4q^2}{4\pi\epsilon_0 a^2}$
 (3) $\left(\frac{1+2\sqrt{2}}{2}\right) \frac{q^2}{4\pi\epsilon_0 a^2}$
 (4) $\left(2+\frac{1}{\sqrt{2}}\right) \frac{q^2}{4\pi\epsilon_0 a^2}$
- Three charges $2q, -q, -q$ are located at the vertices of an equilateral triangle. At the centre of the triangle:-
 (1) The field is zero but potential is non-zero
 (2) The field is non-zero but potential is zero
 (3) Both field and potential are zero
 (4) Both field and potential are non-zero
- Figure shows the electric lines of force emerging from a charged body. If the electric field at A and B are E_A and E_B respectively and if the displacement between A and B is r then:-

 (1) $E_A > E_B$ (2) $E_A < E_B$
 (3) $E_A = \frac{E_B}{r}$ (4) $E_A = \frac{E_B}{r^2}$
- Two spheres A and B of radius 4cm and 6cm are given charges of $80\mu\text{C}$ and $40\mu\text{C}$ respectively. If they are connected by a fine wire, the amount of charge flowing from one to the other is:-
 (1) $20\mu\text{C}$ from A to B
 (2) $16\mu\text{C}$ from A to B
 (3) $32\mu\text{C}$ from B to A
 (4) $32\mu\text{C}$ from A to B
- Three particles, each having a charge of $10\mu\text{C}$ are placed at the corners of an equilateral triangle of side 10cm . The electrostatic potential energy of the system is (Given $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N-m}^2/\text{C}^2$):-
 (1) Zero (2) Infinite
 (3) 27 J (4) 100 J

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7. Two positive charges of 20 coulomb and Q coulomb are situated at a distance of 60 cm. The neutral point between them is at a distance of 20 cm from the 20 coulomb charge. Charge Q is:-

(1) 30 C (2) 40 C
(3) 60 C (4) 80 C

8. Three identical positive point charges, as shown are placed at the vertices of an isosceles right angled triangle. Which of the numbered vectors coincides in direction with the electric field at the mid-point M of the hypotenuse:-



(1) 1 (2) 2
(3) 3 (4) 4

9. The intensity of electric field, due to a uniformly charged infinite cylinder of radius R, at a distance $r(< R)$ from its axis is proportional to

(1) r
(2) r^2
(3) $\frac{1}{r}$
(4) $\frac{1}{r^2}$

10. A thin spherical conducting shell of radius R has a charge q. Another charge Q is placed at the centre of the shell. The electrostatic potential at a point p a distance $\frac{R}{2}$ from the centre of the shell is:-

(1) $\frac{(q+Q)}{4\pi\epsilon_0 R}$ (2) $\frac{2Q}{4\pi\epsilon_0 R}$
(3) $\frac{2Q}{4\pi\epsilon_0 R} - \frac{2q}{4\pi\epsilon_0 R}$ (4) $\frac{2Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 R}$

11. Electric field intensity at a point in between two parallel sheets with like charges of same surface charge densities (σ) is

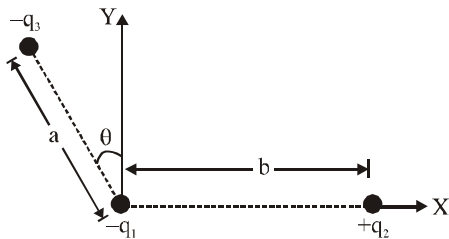
(1) $\frac{\sigma}{2\epsilon_0}$ (2) $\frac{\sigma}{\epsilon_0}$
(3) Zero (4) $\frac{2\sigma}{\epsilon_0}$

12. A metallic wire is folded in the form of a ring of radius 24 cm and is negatively charged to 5×10^{-9} coulomb. The distance from the centre of the ring to the point on the axis of the ring, where the intensity of the electric field is maximum, will be :-

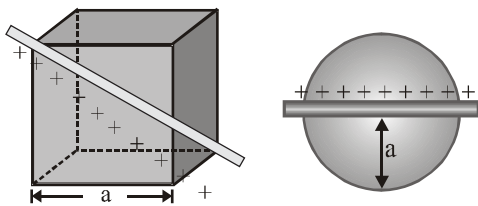
(1) 0.17 m
(2) 1.7 m
(3) 1.7×10^{-2} m
(4) 1.7×10^2 m

SPACE FOR ROUGH WORK

13. Three charge $-q_1$, $+q_2$ and $-q_3$ are placed as shown in the figure. The x-component of the force on $-q_1$ is proportional to:-



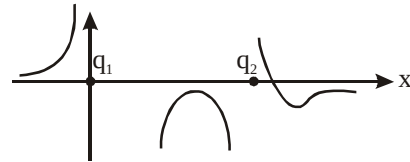
- (1) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \sin \theta$ (2) $\frac{q_2}{b^2} - \frac{q_3}{a^2} \cos \theta$
(3) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \sin \theta$ (4) $\frac{q_2}{b^2} + \frac{q_3}{a^2} \cos \theta$
14. A linear charge having linear charge density λ , penetrates a cube diagonally and then it penetrates a sphere diametrically as shown. What will be the ratio of flux coming out of cube and sphere :-



- (1) $\frac{1}{2}$ (2) $\frac{2}{\sqrt{3}}$ (3) $\frac{\sqrt{3}}{2}$ (4) $\frac{1}{1}$
15. An arc of radius r carries charge. The linear density of charge is λ and the arc subtends an angle $\frac{\pi}{3}$ at the centre. What is electric potential at the centre:-

- (1) $\frac{\lambda}{4\epsilon_0}$ (2) $\frac{\lambda}{8\epsilon_0}$ (3) $\frac{\lambda}{12\epsilon_0}$ (4) $\frac{\lambda}{16\epsilon_0}$

16. Two charge q_1 and q_2 are kept on the x-axis, and the electric field at different points on the x-axis is plotted against x. Choose the correct statement about the nature and magnitude of q_1 and q_2 .



- (1) q_1 is positive, q_2 is negative; $|q_1| > |q_2|$
(2) q_1 is positive, q_2 is negative; $|q_1| < |q_2|$
(3) q_1 is negative, q_2 is positive; $|q_1| > |q_2|$
(4) q_1 is negative, q_2 is positive; $|q_1| < |q_2|$
17. Consider two concentric spherical surfaces S_1 with radius a and S_2 with radius $2a$, both centered at the origin. There is a charge $+q$ at the origin and there are no other charges. Compare the flux ϕ_1 through S_1 with the flux ϕ_2 through S_2 .
- (1) $\phi_1 = 4\phi_2$
(2) $\phi_1 = 2\phi_2$
(3) $\phi_1 = \phi_2$
(4) $\phi_1 = \phi_2 / 2$
18. Intensity of an electric field E due to a dipole, depends on distance r as

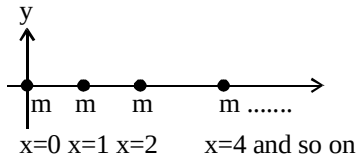
- (1) $E \propto \frac{1}{r^4}$ (2) $E \propto \frac{1}{r^3}$
(3) $E \propto \frac{1}{r^2}$ (4) $E \propto \frac{1}{r}$

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19. If E_a be the electric field strength of a short dipole at a point on its axial line and E_e that on the equatorial line at the same distance, then
 (1) $E_e = 2E_a$
 (2) $E_a = 2E_e$
 (3) $E_a = E_e$
 (4) None of the above
20. The torque acting on a dipole of moment $\vec{P}$ in an electric field $\vec{E}$ is
 (1) $\vec{P} \cdot \vec{E}$ (2) $\vec{P} \times \vec{E}$
 (3) Zero (4) $\vec{E} \times \vec{P}$
21. Electric potential at an equatorial point of a small dipole with dipole moment P (r , distance from the dipole) is :
 (1) Zero (2) $\frac{P}{4\pi\epsilon_0 r^2}$
 (3) $\frac{P}{4\pi\epsilon_0 r^3}$ (4) $\frac{2P}{4\pi\epsilon_0 r^3}$
22. The value of electric potential at any point due to any electric dipole is
 (1) $k \cdot \frac{\vec{p} \times \vec{r}}{r^2}$ (2) $k \cdot \frac{\vec{p} \times \vec{r}}{r^3}$
 (3) $k \cdot \frac{\vec{p} \cdot \vec{r}}{r^2}$ (4) $k \cdot \frac{\vec{p} \cdot \vec{r}}{r^3}$
23. An electric dipole has the magnitude of its charge as q and its dipole moment is $\vec{P}$. It is placed in a uniform electric field $\vec{E}$. If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively
 (1) $2q\vec{E}$ and minimum
 (2) $q\vec{E}$ and $\vec{p} \cdot \vec{E}$
 (3) Zero and minimum
 (4) $q\vec{E}$ and maximum
24. Two spheres of masses m and M are situated in air and the gravitational force between them is F . The space around the masses is now filled with a liquid of specific gravity 3. The gravitational force will now be-
 (1) $3F$ (2) F
 (3) $\frac{F}{3}$ (4) $\frac{F}{9}$
25. Three particles, each of mass M , are situated at the vertices of an equilateral triangle of side ' a '. The only forces acting on the particles are their mutual gravitational forces. It is desired that each particle moves in a circle while maintaining their original separation ' a '. The initial velocity that should be given to each particle and time period of circular motion are respectively-
 (1) $\sqrt{\frac{2GM}{a}}, 2\pi\sqrt{\frac{a^3}{3GM}}$
 (2) $\sqrt{\frac{GM}{a}}, \frac{1}{2\pi}\sqrt{\frac{3GM}{a^3}}$
 (3) $\sqrt{\frac{GM}{a}}, 2\pi\sqrt{\frac{a^3}{3GM}}$
 (4) $\sqrt{\frac{GM}{3a}}, 2\pi\sqrt{\frac{a^3}{3GM}}$

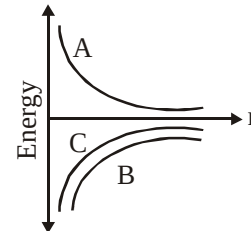
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26. If the earth stops rotating suddenly, the value of g at a place other than poles would :-
 (1) decrease
 (2) remain constant
 (3) increase
 (4) increase or decrease depending on the position of earth in the orbit round the sun
27. A large number of identical point masses m are placed along x -axis at $x = 0, 1, 2, 4, \dots$. The magnitude of gravitational force on mass at origin ($x = 0$), will be :-



- (1) Gm^2 (2) $\frac{4}{3} Gm^2$
 (3) $\frac{2}{3} Gm^2$ (4) $\frac{5}{4} Gm^2$
28. If G is universal gravitation constant and g is acceleration due to gravity, then dimensions of $\frac{G}{g}$ will be :-
 (1) $[M^{-1}L^2]$
 (2) $[M^{-1}L]$
 (3) $[M^{-2}L]$
 (4) $[M^{-1}L^{-2}]$

29. If mass of two point objects are tripled and distance between them is doubled, then gravitational force of attraction between them will :-
 (1) Increase by 225%
 (2) Decrease by 56%
 (3) Increase by 125%
 (4) Decrease by 144%
30. The escape velocity from a planet is v_0 . The escape velocity from a planet having twice the radius but same density will be :-
 (1) $0.5 v_0$ (2) v_0 (3) $2v_0$ (4) $4v_0$
31. The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct



statement about the curves

A, B and C :-

- (1) A shows the kinetic energy, B the total energy and C the potential energy of the system.
 (2) C shows the total energy, B the kinetic energy and A the potential energy of the system.
 (3) C and A are kinetic and potential energies respectively and B is the total energy of the system
 (4) A and B are kinetic and potential energies and C is the total energy of the system.

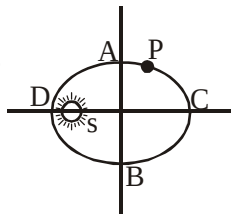
SPACE FOR ROUGH WORK

32. A satellite of mass m , initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is :-

- (1) $\frac{\sqrt{3}}{4}mgR$
- (2) $\frac{1}{2}mgR$
- (3) $\frac{1}{4}mgR$
- (4) $\frac{3}{4}mgR$

Paragraph for question no. 7 and 8 :

Figure shows the orbit of a planet P round the sun S . AB and CD are the minor and major axes of the ellipse.



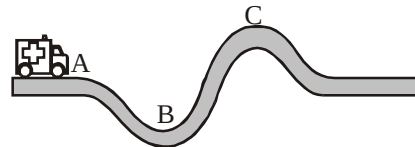
33. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA , then :-
- (1) $t_1 = t_2$
 - (2) $t_1 > t_2$
 - (3) $t_1 < t_2$
 - (4) Nothing can be concluded
34. If U is the potential energy and K kinetic energy then $|U| > |K|$ at :-
- (1) Only D
 - (2) Only C
 - (3) Both D and C
 - (4) Neither D nor C

35. If a_r and a_t represents radial and tangential acceleration, the motion of a particle will be circular if :

- (i) $a_r = 0$ and $a_t = 0$
- (ii) $a_r = 0$ but $a_t \neq 0$
- (iii) $a_r \neq 0$ but $a_t = 0$
- (iv) $a_r \neq 0$ and $a_t \neq 0$

- (1) (i, iii)
- (2) (ii, iii)
- (3) (iii, iv)
- (4) (ii, iv)

36. A vehicle is moving on a track as shown in Fig. The weight of the vehicle is :



- (1) maximum at A
- (2) maximum at B
- (3) maximum at C
- (4) same at A, B and C

37. A 2 kg stone at the end of a string 1 m long is whirled in a vertical circle. At some point its speed is 4m/s. The tension of the string is 51.6 newton. At this instant the stone is :

- (1) at the top of the circle
- (2) at the bottom of the circle
- (3) half way down
- (4) none of these

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38. The potential energy of a particle of mass 1 kg moving along x-axis given by

$$U(x) = \left[\frac{x^2}{2} - x \right] \text{ J. If total mechanical}$$

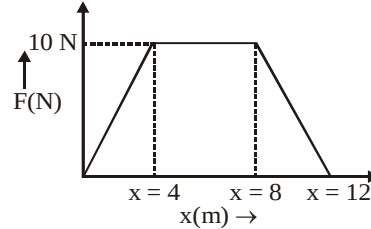
energy of particle is 2J then find its maximum speed (in m/s) :-

- (1) $\sqrt{5}$
- (2) $\sqrt{7}$
- (3) $\sqrt{3}$
- (4) None

39. A man pulls a bucket full of water from h metre deep well. If the mass of rope is m and mass of bucket full of water is M, then work done by the man is :-

- (1) $\left(\frac{M}{2} + m \right) gh$
- (2) $\left(\frac{M+m}{2} \right) gh$
- (3) $\left(M + \frac{m}{2} \right) gh$
- (4) $(M + m) gh$

40. A particle of mass 0.1 kg is subjected to a force which varies with distance as shown. If it starts its journey from rest at $x = 0$, then its velocity at $x = 12$ m is :-



- (1) 0 m/s
 - (2) $20\sqrt{2}$ m/s
 - (3) $20\sqrt{3}$ m/s
 - (4) 40 m/s
41. The rate of doing work by force acting on a particle moving along x-axis depends on position x of particle and is equal to $2x$. The velocity of particle is given by expression :-

- (1) $\left[\frac{3x^2}{m} \right]^{1/3}$
- (2) $\left[\frac{3x^2}{2m} \right]^{1/3}$
- (3) $\left(\frac{2mx}{9} \right)^{1/2}$
- (4) $\left[\frac{mx^2}{3} \right]^{1/2}$

SPACE FOR ROUGH WORK

42. A body of mass m accelerates uniformly from rest to velocity v_1 in time interval T_1 . The instantaneous power delivered to the body as a function of time t is :-

- (1) $\frac{mv_1^2}{T_1^2} t$
(2) $\frac{mv_1}{T_1^2} t$
(3) $\left(\frac{mv_1}{T_1}\right)^2 t$
(4) $\frac{mv_1^2}{T_1} t^2$

43. The position (x) of a body moving along x -axis at time (t) is given by $x = 3t^2$, where x is in metre and t is in second. If mass of body is 2 kg, then find the instantaneous power delivered to body by force acting on it at $t = 4$ s :-

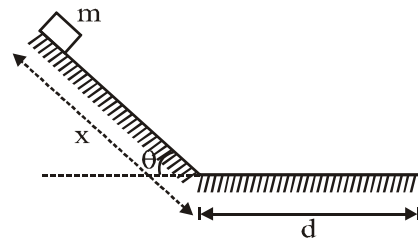
- (1) 288 W
(2) 280 W
(3) 290 W
(4) None

44. The velocity (v) of a particle of mass m moving along x -axis is given by $v = \alpha\sqrt{x}$, where α is a constant. Find work done by force acting on particle during its motion from $x = 0$ to $x = 2m$:-

- (1) $m\alpha^2$
(2) $m\alpha$
(3) $\frac{m\alpha}{2}$

- (4) None

45. A block of mass m is released on the top of a smooth inclined plane of length x and inclination θ as shown in figure. Horizontal surface is rough. If block comes to rest after moving a distance d on the horizontal surface, then coefficient of friction between block and surface is :-



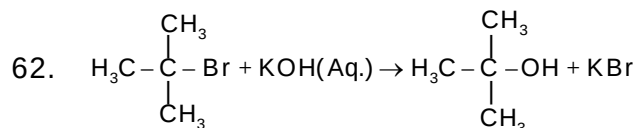
- (1) $\frac{x \sin \theta}{2d}$
(2) $\frac{x \cos \theta}{2d}$
(3) $\frac{x \sin \theta}{d}$
(4) $\frac{x \cos \theta}{d}$

SPACE FOR ROUGH WORK

46. A chemical reaction is at equilibrium when
- (1) Reactants are completely transformed into products
 - (2) The rates of forward and backward reactions are equal
 - (3) Formation of products is minimised
 - (4) Equal amounts of reactants and products are present
47. In a reaction the rate of reaction is proportional to its active mass, this statement is known as
- (1) Law of mass action
 - (2) Le-chatelier principle
 - (3) Faraday law of electrolysis
 - (4) Law of constant proportion
48. For the system $3A + 2B \rightleftharpoons C$, the expression for equilibrium constant is
- (1) $\frac{[3A][2B]}{[C]}$
 - (2) $\frac{[C]}{[3A][2B]}$
 - (3) $\frac{[A]^3[B]^2}{[C]}$
 - (4) $\frac{[C]}{[A]^3[B]^2}$
49. Which of the following is a non-electrolyte
- (1) NaCl
 - (2) $CaCl_2$
 - (3) $C_{12}H_{22}O_{11}$
 - (4) CH_3COOH
50. For a weak acid HA Ostwald's dilution law is represented by the equation
- (1) $K_a = \frac{\alpha c}{1 - \alpha^2}$
 - (2) $K_a = \frac{\alpha^2 c}{1 - \alpha}$
 - (3) $\alpha = \frac{K_a c}{1 - c}$
 - (4) $K_a = \frac{\alpha^2 c}{1 - \alpha^2}$
51. Degree of dissociation of 0.1N CH_3COOH is (Dissociation constant = 1×10^{-5})
- (1) 10^{-5}
 - (2) 10^{-4}
 - (3) 10^{-3}
 - (4) 10^{-2}
52. The conjugate acid of HPO_3^{2-} is
- (1) H_3PO_4
 - (2) H_3PO_3
 - (3) $H_2PO_3^-$
 - (4) PO_4^{3-}
53. The solubility product of a salt having general formula MX_2 in water is : 4×10^{-12} . The concentration of M^{2+} ions in the aqueous solution of the salt is
- (1) $2.0 \times 10^{-6} M$
 - (2) $1.0 \times 10^{-4} M$
 - (3) $1.6 \times 10^{-4} M$
 - (4) $4.0 \times 10^{-10} M$
54. A buffer solution with pH = 9 is to be prepared by mixing NH_4Cl and NH_4OH . Calculate the number of moles of NH_4Cl that should be added to one litre of 1.0 M NH_4OH . [$K_b(NH_4OH) = 1.8 \times 10^{-5}$]
- (1) 3.4
 - (2) 2.6
 - (3) 1.5
 - (4) 1.8
55. Which of the following chemicals, in addition to water, are used for the manufacture of Na_2CO_3 by Solvay process
- (1) NaCl, CO and NH_3
 - (2) NaCl, CO_2 and NH_3
 - (3) NaCl, NH_4Cl and CO_2
 - (4) $NaHCO_3$, CO and NH_3
56. Gypsum $CaSO_4 \cdot 2H_2O$ on heating to about $120^\circ C$ forms a compound which has the chemical composition represented by
- (1) $CaSO_4$
 - (2) $2CaSO_4 \cdot H_2O$
 - (3) $CaSO_4 \cdot H_2O$
 - (4) $2CaSO_4 \cdot 3H_2O$

SPACE FOR ROUGH WORK

57. The enolic form of acetone contains
 (1) 8 σ -bonds, 2 π -bonds and 1 lone pairs
 (2) 9 σ -bonds, 1 π -bond and 2 lone pairs
 (3) 9 σ -bonds, 2 π -bonds and 1 lone pairs
 (4) 10 σ -bonds, 1 π -bonds and 1 lone pairs
58. Which one of the following orders is correct regarding the inductive effect of the substituents
 (1) $-\text{NR}_2 < -\text{OR} > -\text{F}$
 (2) $-\text{NR}_2 > -\text{OR} > -\text{F}$
 (3) $-\text{NR}_2 < -\text{OR} < -\text{F}$
 (4) $-\text{NR}_2 > -\text{OR} < -\text{F}$
59. Reactivity towards nucleophilic addition reaction of (I) HCHO , (II) CH_3CHO , (III) CH_3COCH_3 is
 (1) $\text{II} > \text{III} > \text{I}$ (2) $\text{III} > \text{II} > \text{I}$
 (3) $\text{I} > \text{II} > \text{III}$ (4) $\text{I} > \text{II} < \text{III}$
60. Which of the following reaction requires radical as intermediate
 (1) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \rightarrow \text{CH}_3 - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$
 (2) $\text{CH}_3 - \text{CHO} + \text{HCN} \rightarrow \text{CH}_3 - \underset{\text{OH}}{\overset{\text{CN}}{\text{CH}}}$
 (3) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \xrightarrow{\text{peroxide}} \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$
 (4) $\text{CH}_3\text{CHO} + \text{NH}_2\text{OH} \xrightarrow{\text{H}^+} \text{CH}_3 - \text{CH} = \text{N} - \text{OH}$
61. Which of the following species is paramagnetic in nature
 (1) Free radical
 (2) Carbonium ion
 (3) Carbanion
 (4) All the above



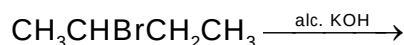
above reaction is

- (1) SN^1
 (2) SN^2
 (3) E_1
 (4) Both (1) and (3)
63. The reaction

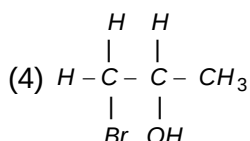
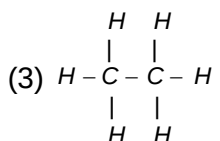
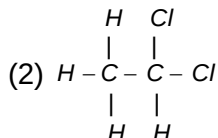
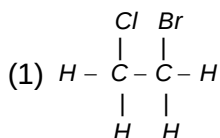
$$\text{R} - \underset{\text{X}}{\overset{\text{O}}{\text{C}}} + \text{Nu}^- \rightarrow \text{R} - \underset{\text{Nu}}{\overset{\text{O}}{\text{C}}} + \text{X}^-$$
, is fastest
 when X is
 (1) Cl
 (2) NH_2
 (3) OC_2H_5
 (4) OCOR
64. Examine the following statements pertaining to an SN^2 reaction
 (1) The rate of reaction is independent of the concentration of the nucleophile
 (2) The nucleophile attacks the C atom on the side of the molecule opposite to the group being displaced
 (3) The reaction proceeds with simultaneous bond formation and bond rupture/cleavage
- Amongst the following which of the above are true
 (1) 1, 2
 (2) 1, 3
 (3) 1, 2, 3
 (4) 2, 3

SPACE FOR ROUGH WORK

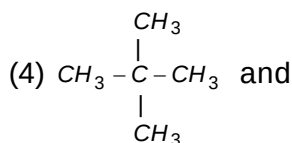
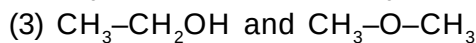
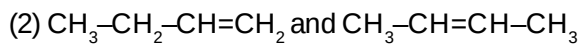
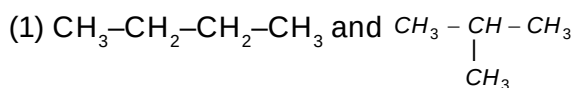
65. Which of the following applies in the reaction,



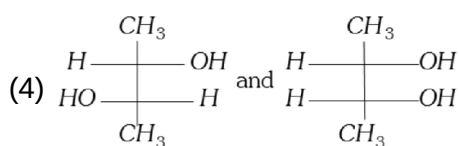
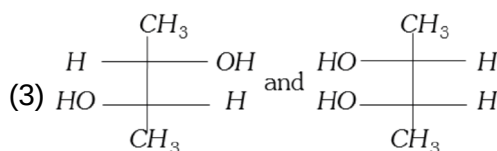
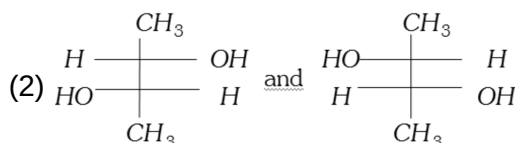
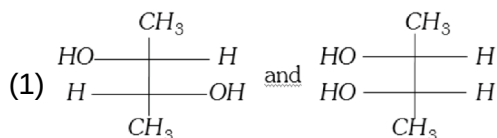
- (i) $\text{CH}_3\text{CH}=\text{CHCH}_3$ (major product)
 (ii) $\text{CH}_2=\text{CHCH}_2\text{CH}_3$ (minor product)
- (1) Markovnikov's rule
 (2) Saytzeff's rule
 (3) Kharasch effect
 (4) Hofmann's rule
66. Which of the following cannot undergo nucleophilic substitution under ordinary conditions
- (1) Chlorobenzene
 (2) *tert*-butylchloride
 (3) Isopropyl chloride
 (4) None of these
67. Which one of the following contains asymmetric carbon atom



68. Which of the following pairs is an example of position isomerism

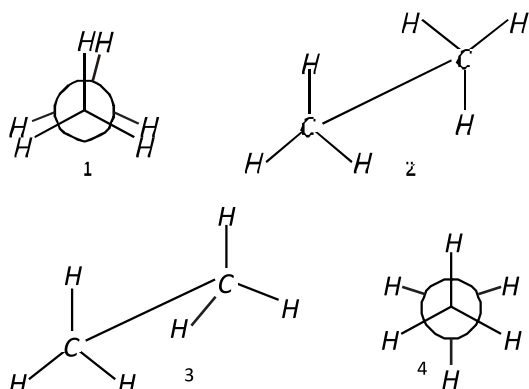


69. Geometrical isomerism is shown by
- (1) 2-butene (2) 2-butyne
 (3) 2-butanol (4) Butanal
70. Which of the following pairs of compounds are enantiomers



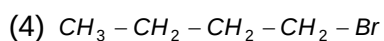
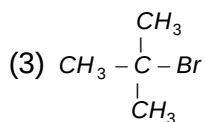
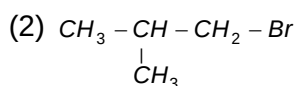
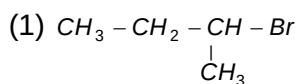
SPACE FOR ROUGH WORK

71. Which are the staggered forms of ethane



- (1) 1 and 4 (2) 3 and 4
 (3) 2 and 4 (4) 1 and 3

72. What is the chief product obtained when *n*-butane is treated with bromine in the presence of light at 130°C



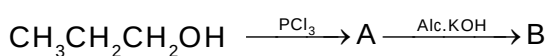
73. $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \xrightarrow{\text{CCl}_4} \text{(A)}$ will be :-

- (1) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$
 (2) $\text{CH}_3 - \text{CHBr} - \text{CH}_3$
 (3) $\text{BrCH}_2 - \text{CH} = \text{CH}_2$
 (4) $\text{CH}_2 = \text{C} = \text{CH}_2$

74. The product of reaction between propene and HBr in the presence of a peroxide is

- (1) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Br}$
 (2) $\text{CH}_3 - \text{CHBr} - \text{CH}_3$
 (3) $\text{CH}_3 - \text{CH}_2\text{Br}$
 (4) $\text{CH}_3 - \text{CH} = \text{CHBr}$

75. The compound B formed in the following sequences of reactions is



- (1) Propyne
 (2) Propene
 (3) Propanol
 (4) Propane

76. *n*-propyl bromide on treatment with ethanolic potassium hydroxide produces

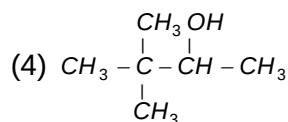
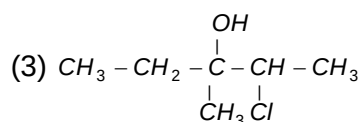
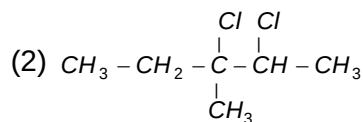
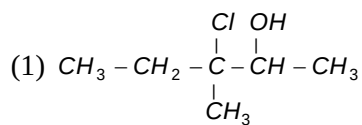
- (1) Propane
 (2) Propene
 (3) Propyne
 (4) Propanol

77. $\text{CH}_2 = \text{CHCl}$ reacts with *HCl* to form

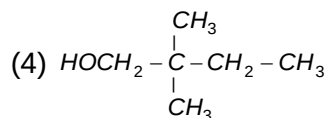
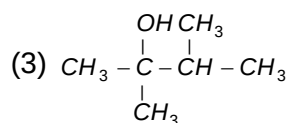
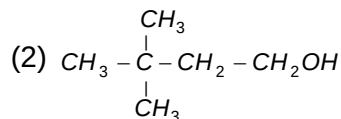
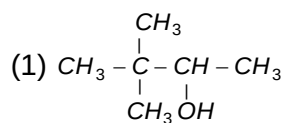
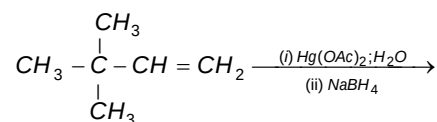
- (1) $\text{CH}_2\text{Cl} - \text{CH}_2\text{Cl}$
 (2) $\text{CH}_3 - \text{CHCl}_2$
 (3) $\text{CH}_2 = \text{CHCl} \cdot \text{HCl}$
 (4) None of these

SPACE FOR ROUGH WORK

78. The predominant product formed, when 3-methyl-2-pentene reacts with HOCl , is

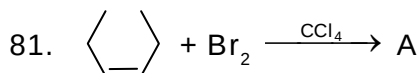


79. The product of following reaction is

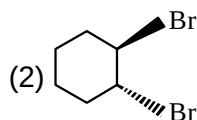
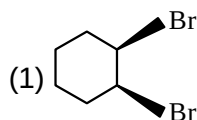


80. When acetylene is passed into dilute sulphuric acid containing Hg^{2+} ions, the product formed is

- (1) Acetone
- (2) Acetic acid
- (3) Acetaldehyde
- (4) Formaldehyde



A will have configuration -

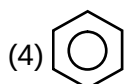
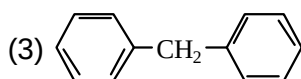
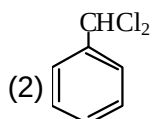
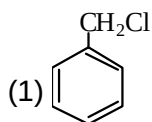
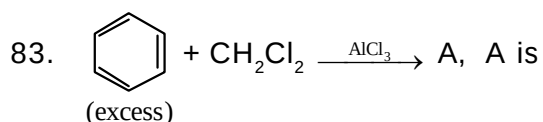


- (3) both true
- (4) none is true

82. In the chlorination of $(\text{CH}_3)_2\text{CH} - \text{CH}_2 - \text{CH}_3$ the substitution at :

- (1) 1° carbon would be fastest
- (2) 2° carbon would be fastest
- (3) 3° carbon would be fastest
- (4) 1° , 2° , 3° carbon atoms all will occur at the same rate

SPACE FOR ROUGH WORK



84. Which of the following is the strongest o, p-directing group ?

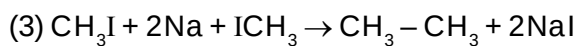
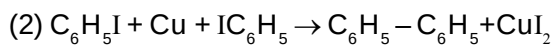
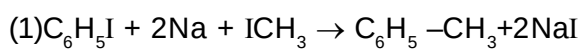
(1) OH

(2) Cl

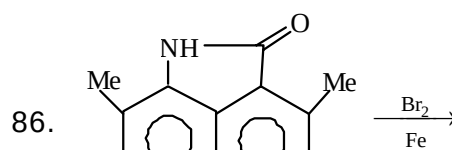
(3) Br

(4) C_6H_5

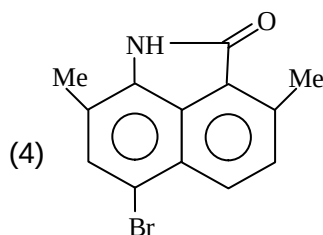
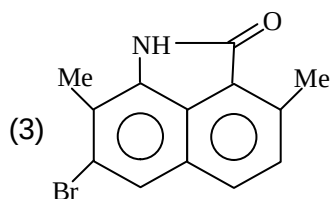
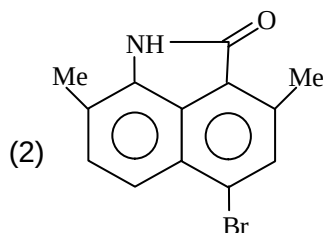
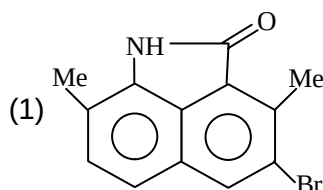
85. Which of the following is Wurtz – Fittig reaction–



(4) None of the above



Major product of above reaction is:



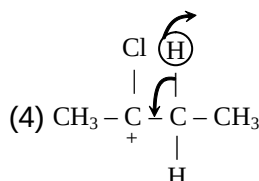
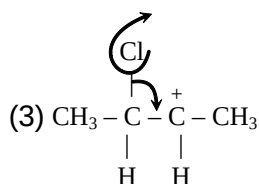
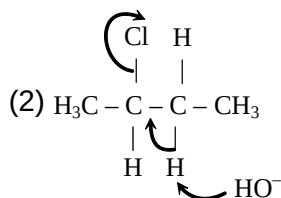
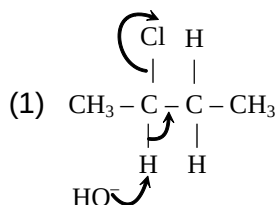
SPACE FOR ROUGH WORK

87. Which of the following is the correct order of decreasing S_N2 reactivity ?

- (1) $RCH_2X > R_3CX > R_2CHX$
- (2) $RCH_2X > R_2CHX > R_3CX$
- (3) $R_3CX > R_2CHX > RCH_2X$
- (4) $R_2CHX > R_3CX > RCH_2X$

(X = a halogen)

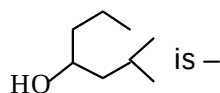
88. For a compound undergoing E_2 reaction which step is correct -



89. Which of the following alkyl halides on conversion into the Grignard reagent followed by treatment with water would yield 2-methylbutane?

- (1) 2-Chloro-2-methylbutane
- (2) 1-Chloro-2-methylbutane
- (3) 2-Chloro-3-methylbutane
- (4) All of them

90. The IUPAC name of the compound



- (1) 3,3 -dimethyl-1-hydroxy cyclohexane
- (2) 1, 1 – dimethyl–3– hydroxy cyclohexane
- (3) 3,3-dimethyl-1-cyclohexanol
- (4) 1,1 - dimethyl - 3- cyclohexanol

SPACE FOR ROUGH WORK

91. "Gametes are never hybrids", this is a statement of law of:
(1) Dominance
(2) Segregation
(3) Polygenes
(4) Independent assortment
92. A self fertilising trihybrid plant forms:
(1) 4 different gametes and 64 different zygote.
(2) 8 different gametes and 16 different zygote
(3) 8 different gamete and 64 different zygote
(4) 8 different gametes and 32 different zygote
93. What is the probability of a gamete having recessive alleles only, if genotype of the plant is Aa bb Cc?
(1) 50% (2) 25% (3) 100% (4) 75%
94. Mendel's experimental plant was *Pisum sativum*, but he failed to find similar results on :
(1) *Hieracium* (2) *Raphanus sativa*
(3) Sweet pea (4) Both (1) and (2)
95. Human blood grouping is common example of:
(i) Quantitative inheritance
(ii) Multiple allelism
(iii) Incomplete dominance
(iv) Co - Dominance
(1) Only (ii) (2) Only (iv)
(3) Both (ii) and (iii) (4) Both (ii) and (iv)
96. Which of the following is not a codominant trait?
(1) AB blood group
(2) MN blood group
(3) Starch grain size in pea
(4) Sickle cell anaemia ($Hb^A Hb^S$)
97. Types of gametes produced by a diploid plant with genotype Cc Dd Ee FFGg hh:
(1) 16 (2) 8 (3) 4 (4) 32
98. Proportion of offspring with genotype aa Bb Cc Dd in a cross Aa BB Cc DD $\times$ aa bb Cc Dd will be:
(1) 1/16 (2) 4/16 (3) 1/8 (4) 8/16
99. How many F_2 plants having minimum three dominant allele in a typical dihybrid cross?
(1) 5/16 (2) 9/16
(3) 4/16 (4) 10/16
100. Select the parents genotype that cannot produce child with blood group 'O'.
(1) $I^A i^\circ \times I^A i^\circ$ (2) $I^B i^\circ \times I^B i^\circ$
(3) $I^A I^A \times I^B i^\circ$ (4) $i^\circ i^\circ \times i^\circ i^\circ$
101. Select the incorrect match pair w.r.t. garden pea.
(1) Aa $\times$ Aa $\longrightarrow$ Selfing
(2) Aa Bb $\times$ aa bb $\longrightarrow$ Dihybrid test cross
(3) Aa $\times$ aa $\longrightarrow$ Out cross
(4) AA $\times$ AA $\longrightarrow$ Self pollination
102. Consider the cross for the pleiotropic gene (R) in pea seed

Rr	Rr
----	----

(Intermediate starch grain) $\times$ (Intermediate starch grain)
Accordingly which of the following is true for F_2 generation :
(1) It do not possesses intermediate sized starch grains
(2) It do not possesses round seeds
(3) Incomplete dominance can be observed
(4) More than one option is correct
103. A plant having genotype AABBCC is 36 cm long and another plant having genotype aa bb cc is 12 cm long. What will be the length of plant having genotype AABbcc?
(1) 28 cm (2) 24 cm
(3) 10 cm (4) 22 cm
104. Choose wrong statement for chromosomal theory of inheritance.
(1) Genes and chromosomes both occur in pair
(2) Proposed by Sutton and Boveri
(3) Behaviour of chromosome movement was studied in mitosis
(4) Independent pair of genes and chromosomes segregate independently of each other

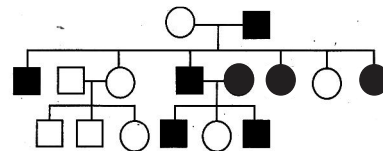
105. Henking first observed sex chromosome in the form of :-
 (1) Z-chromosome (2) X-Body
 (3) Y-Chromosome (4) Barr body
106. In a complementary gene interaction, calculate the number of phenotype and genotype produced in a cross $Aa Bb \times aa BB$
 (1) 1 Phenotype, 2 genotype
 (2) 2 Phenotype, 2 genotype
 (3) 2 Phenotype, 4 genotype
 (4) 2 Phenotype, 3 genotype
107. A holandric gene causes Hypertrichosis. When a man with hairy pinna marries a color blind women, what percentage of their son would be expected to have color blindness?
 (1) 50% (2) 25% (3) 75% (4) 100%
108. In Grasshoppers, females are
 (1) ZZ (2) XX (3) ZO (4) XO
109. Heterogametic male is found in all except
 (1) Mammals (2) *Drosophila*
 (3) Birds (4) Grasshopper
110. *Dioscorea* and *Vallisneria* show which type of sex determination?
 (1) XX-XY type (2) XX-XO type
 (3) ZZ-ZW type (4) ZZ-ZO type
111. Number of Barr bodies in XXX female would be:
 (1) 4 (2) 2 (3) 3 (4) 1
112. Foetal sex can be determined from cells present in amniotic fluid by looking for:
 (1) Auto somes
 (2) Marker chromosome
 (3) Barr bodies
 (4) Chiasmata
113. Chromosome number of female and male honey bee is:
 (1) 32, 16 (2) 16, 8
 (3) 20, 24 (4) 32, 32
114. In a cross performed by Morgan in *Drosophila*, the number of offsprings obtained were as follows :
 $W^+ m^+ = 75$, $Wm = 75$
 $+m = 25$, $w + = 25$
 Find out the distance between gene W and m.
 (1) 37.2 (2) 18.6 (3) 62.8 (4) 25

115. In honey bee, male do not have and cannot have, but have a and
- (1) Father, sons, grandfather, grandsons
 (2) Grandfather, grandsons, father, sons
 (3) Sons, grandsons, father, grandfather
 (4) Grandsons, sons, father, grandfather
116. Method of sex determination in *Drosophila* and birds are:
 (1) XX-XO and XX-XY respectively
 (2) XX - XY and ZZ - ZW respectively
 (3) XX - XY and ZZ - ZO respectively
 (4) XX-XO and ZZ-ZO respectively

117. In Pedigree analysis the symbol  represents :

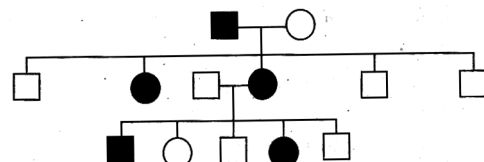
- (1) Consanguineous marriage
 (2) Sex unspecified
 (3) Monozygotic twins
 (4) Dizygotic twins

118. Identify the type of inheritance in the given pedigree.



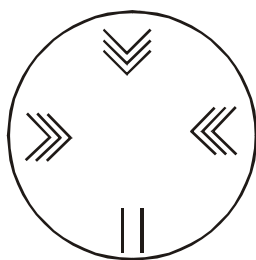
- (1) X-linked Dominant
 (2) X-linked recessive
 (3) X-Y linked
 (4) Autosomal dominant

119. In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.



- (1) Recessive Y - linked
 (2) Recessive X-linked
 (3) Dominant Y - linked
 (4) Dominant X - linked

120. Which of the following most appropriately describes haemophilia?
- Autosomal recessive gene disorder
 - X-linked recessive gene disorder
 - Autosomal disorder
 - X-linked dominant gene disorder
121. In a random mating population of 200 individual 128 have AA, 64 have Aa, 8 have 'aa' genotype. Find out frequency of dominant and recessive allele respectively.
- 0.8, 0.2
 - 0.9, 0.1
 - 0.10, 0.02
 - 0.2, 0.8
122. In a large population frequency of recessive allele is 60%. Find out frequency of homozygous individual
- 0.10
 - 0.6
 - 0.52
 - 0.9
123. In a random mating population of 5000 individual, frequency of dominant organism are 0.51. Find out number of homozygous individual :-
- 2100
 - 2900
 - 3100
 - 2982
124. Recombination frequency between linked gene are as follows?
- A & B – 40, A & C – 10, A & D – 15,
C & D – 5, C & B – 30, D & B – 25.
- CDAB
 - DBCA
 - ABCD
 - ACDB
125. Karyotype of drosophila given below then find out sex ratio of drosophila



- 2.5, supermale
 - 1.5, supermale
 - 0.66, intersex
 - 0.5, male
126. In slime moulds spores possess true cell wall which is made up of
- Cellulose
 - Chitin
 - Lignin
 - Suberin

127. Paramylon is the reserve food in
- Diatoms
 - Dinoflagellates
 - Euglenoids
 - Slime moulds
128. In which of the following protist, the presence of sunlight changes the mode of nutrition?
- Physarum*
 - Gonyaulax*
 - Desmides*
 - Euglena*
129. Find an incorrect match.
- Lichen - Pollution indicator
 - Euglena* - Sleeper animalcule
 - Slime moulds - plasmodium
 - Dinoflagellates - Whirling whips
130. Cellulosic cell wall, Coenocytic mycelium aquatic habitat are present in members of class
- Phycomycetes
 - Ascomycetes
 - Deuteromycetes
 - Basidiomycetes
131. Fungi are filamentous with exception of
- Mucor*
 - Puccinia*
 - Saccharomyces*
 - Aspergillus*
132. Late blight of potato caused by
- Phytophthora*
 - Puccinia*
 - Alternaria*
 - Ustilago*
133. In class Phycomycetes asexual spores are
- Zoospore and Oospore
 - Aplanospore and Zoospore
 - Conidia and Zygosporangium
 - Zygosporangium and Oospore
134. Yeast commonly grown in a medium rich in
- Protein
 - Sugar
 - Lipids
 - Vitamins
135. Clamp connection during sexual reproduction is a characteristic feature of
- Sac fungi
 - Club fungi
 - Conjugation fungi
 - Algal fungi

136. At synaptic cleft neurotransmitter acetylcholine is disintegrated by the enzyme :-

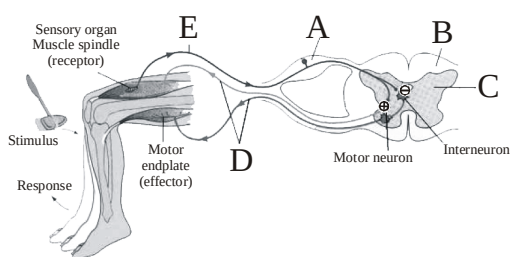
- (1) Hydrolases
- (2) Acetylcholinesterase
- (3) Monocarboxylase
- (4) ATPase

137. Which of the following statements is false about the electrical synapse ?

- I. At electrical synapses, the membranes of pre and post synaptic neurons are in very close proximity.
- II. Electrical current can flow directly from one neuron into the other across the synapses.
- III. Transmission of an impulse across electrical synapses is very similar to impulse conduction along single axon.
- IV. Electrical synapses pass electrical signal between cells with the use of Ach
- V. Electrical synapses are fast.
- VI. Electrical synapses are rare in our system.

- (1) I and II
- (2) Only II
- (3) Only IV
- (4) Only V

138. The following diagrammatic representation of reflex action shows knee jerk reflex.



In which of the following options corrects words for all the 5 blanks (A to E) are indicated?

	A	B	C	D	E
(1)	Dorsal root ganglion	White matter	Gray matter	Afferent pathway	Efferent pathway
(2)	Dorsal root ganglion	White matter	Gray matter	Efferent pathway	Afferent pathway
(3)	Dorsal root ganglion	Gray matter	White matter	Efferent pathway	Afferent pathway
(4)	Ventral root ganglion	White matter	Gray matter	Efferent pathway	Afferent pathway

139. Which is not the property of conditioned reflex?

- (1) Stable
- (2) Non-genetic
- (3) Complex
- (4) Unstable

140. Which pair is not matched correctly ?

- (1) Ramus dorsalis – Sympathetic nervous system
- (2) Largest cranial nerve – trigeminal nerve
- (3) Longest cranial nerve – Vagus nerve
- (4) Ramus ventralis – Somatic nervous system

141. Which of the following cranial nerve originates from pons and receives taste from tongue :-

- (1) 7th nerve
- (2) 9th nerve
- (3) 5th nerve
- (4) 12th nerve

142. Which of the following are the functions of medulla oblongata ?

- (1) Cardiovascular reflex
- (2) Gastric secretion
- (3) Respiration
- (4) All of the above

143. Which part of brain receives and integrates visual, tactile and auditory inputs ?

- (1) Cerebellum
- (2) Midbrain
- (3) Pons
- (4) Medulla oblongata

144. Which of the following part of brain has very convoluted surface in order to provide additional space for many more neurons ?

- (1) Pons
- (2) Medulla oblongata
- (3) Cerebellum
- (4) Midbrain

145. In which stage, ion channel and Na⁺ VGC are functioning but K⁺VGC and Na⁺K⁺ pump are not functioning :-

- (1) Polarisation
- (2) Depolarisation
- (3) Repolarisation
- (4) Hyperpolarisation

146. Which pair is not matched correctly ?

- (1) Microgliaocytes – Ectodermal
- (2) Astrocytes – Blood Brain Barrier
- (3) Schwann cells – Formation of myelin sheath in PNS
- (4) Oligodendrocytes – Formation of myelin sheath in CNS

147. Find out the correct match from the following table :-

Column-I	Column-II	Column-III
(i) Luteal phase	Duration 15 th to 28 th day	Corpus luteum attains maximum size on 22 th day
(ii) Proliferative phase	Duration 6 th to 13 th day	On 14 th day ovulation occurs and corpus luteum develops
(iii) Follicular phase	Duration 16-17 day	Corpus luteum degenerates

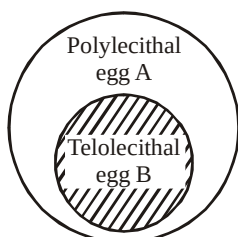
- (1) (i) only
- (2) (i) and (ii)
- (3) (iii) only
- (4) (ii) and (iii)

148. Find out the correct match from the following table.

Column-I	Column-II	Column-III
(i) Alecithal	Yolk absent	Human
(ii) Microlecithal	Yolk very small	Eutheria
(iii) Mesolecithal	Moderate yolk	Amphibia

- (1) (i) & (ii)
- (2) (ii) & (iii)
- (3) (i) & (iii)
- (4) (i), (ii), (iii)

149.



In above diagram frog egg and bird egg can be placed respectively in :-

- (1) Circle 'A' and 'A'
- (2) Circle A and B
- (3) Circle B and B
- (4) Circle B and A

150. Find out the correct match from the table :-

Column-I	Column-II	Column-III
(i) Tertiary	Oocyte	Corona radiata
(ii) Secondary	Ovary	Chorion
(iii) Primary	Oviduct	Zona radiata

- (1) (ii) only
- (2) (i) & (ii)
- (3) (ii) & (iii)
- (4) (i) & (iii)

151. Consider the following four statements (a-d) and select option which includes all the correct ones only.

- (a) The reproductive cycle in the female primates is called menstrual cycle
- (b) Second polar body has 22 + X chromosome
- (c) Antrum is found in secondary follicle
- (d) For normal fertility, at least 60% sperms must have normal shape and size

Options :-

- (1) a, b, c
- (2) a, b, d
- (3) a, b, c, d
- (4) b, c, d

152. Consider the following four statements (a-d) and select the option which includes all the correct ones only.

- (a) GnRH is a hypothalamic hormone
- (b) LH acts at the leydig cells
- (c) Androgen stimulate the process of oogenesis
- (d) FSH acts on the sertoli cells

Options :-

- (1) a, b, d
- (2) a, b, c, d
- (3) b, c, d
- (4) a, d

153. Consider the following four statements (a-d) and select the option which includes all the correct ones only :-

- (a) The primary oocyte forms a new membrane called zona pellucida surrounding it.
- (b) The graafian follicle further changes into the tertiary follicle
- (c) No more oogonia are formed and added after birth
- (d) Many oogonia are formed after birth

Options :-

- (1) a, b, c
- (2) a, c
- (3) a, d
- (4) a, b, d

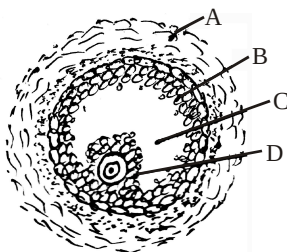
154. Select wrong statement :-

- (1) One ovum is released during the middle of each menstrual cycle
- (2) Menstruation only occurs if the released ovum is not fertilized
- (3) Lack of menstruation always will be indicator of pregnancy
- (4) The secretion of gonadotropins increases gradually during the follicular phase

155. Which match is not correct ?

- (1) Cleidoic egg - birds
- (2) Centrolecithal egg - Reptile
- (3) Telolecithal - Amphibia
- (4) Discoidal - Prototherian

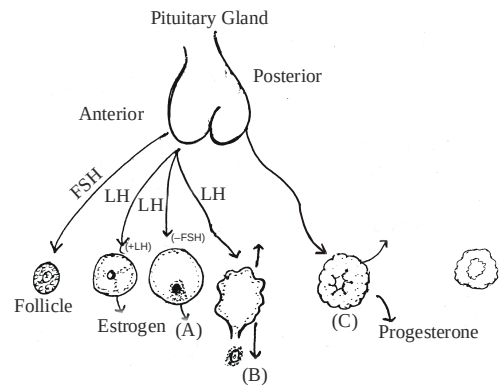
156.



In above diagram sketch of a mature graafian follicle is given. Identify the parts labelled A, B, C and D and select the right option about them.

	Part-A	Part-B	Part-C	Part-D
(1)	Granulosa cells	Corona radiata	Antrum	Stroma cells
(2)	Theca externa	Oogonia	Sertoli cells	Corona radiata
(3)	Theca externa	Granulosa cells	Liquor folliculi	Corona radiata
(4)	Corona radiata	Theca interna	Oogonia	Zona pellucida

157.



See the diagram carefully and choose correct option.

	A	B	C
(1)	Progesterone	Menstruation	Corpus albicans
(2)	Estrogen	Ovulation	Corpus luteum
(3)	Relaxin	Secondary oocyte	Corpus luteum
(4)	Estrogen	Second polar body	Cumulus ooporus

158. Which of the following statements are true about cleavage ?

- (a) K.I. increases during division
 - (b) Both blastomeres and embryo size are reduced
 - (c) Interphase stage only shows, s-phase
 - (d) Yolk is inversely proportional to division
- (1) a, b, c, d
 - (2) a, c, d
 - (3) b, c, d
 - (4) Only c and d

159. Which type of cleavage is present in alecithal egg of human ?
- (1) Determinate and equal holoblastic
 - (2) Indeterminate and unequal holoblastic
 - (3) Determinate and unequal meroblastic
 - (4) Indeterminate and equal meroblastic
160. Which statement is wrong about human embryology ?
- (1) Zona pellucida disappears from blastocyst
 - (2) Implantation occurs in Gastrula stage
 - (3) Interstitial type of implantation
 - (4) 8 weeks embryo is called foetus
161. What correct sequence is followed in developing embryo ?
- (1) Ectoderm → Endoderm → Mesoderm
 - (2) Endoderm → Ectoderm → Extraembryonic mesoderm → Intraembryonic mesoderm
 - (3) Ectoderm → Intraembryonic mesoderm → Endoderm
 - (4) Chorion → Amnion → Endoderm → Ectoderm → Mesoderm
162. Gastrulation is characterised by :-
- (1) Morphogenetic movement of blastomeres
 - (2) Formation of three germinal layer
 - (3) (1) and (2) both
 - (4) Neural tube formation
163. Which of the following statements are true about the female reproductive system ?
- (a) Both estrogen and progesterone are necessary for ovulation
 - (b) Progesterone production is under the control of LH
 - (c) The part of the oviduct is isthmus, has a narrow lumen and it joins the Uterus
 - (d) The middle thick layer of Uterus (myometrium) is made up of skeletal muscles
 - (e) Estrogen indirectly tends to inhibit the production of FSH
- (1) a, b, c are true
- (2) a, b, d are true
- (3) b, c, e are true
- (4) b, c, d, e are true
164. If we could monitor the amount of total gonadotropin activity in pregnant women, we would expect :-
- (1) High levels of FSH and LH in the uterus to stimulate endometrial thickening
 - (2) High levels of circulating FSH and LH to stimulate implantation of the embryo
 - (3) High levels of HCG in the uterus to stimulate endometrial thickening
 - (4) High levels of circulating HCG to stimulate estrogen and progesterone synthesis
165. Polar bodies are produced during the formation of :
- (1) Sperm
 - (2) Primary oocyte
 - (3) Spermatocytes
 - (4) Secondary oocyte
166. Which hormone is secreted by hypothalamus as well as placenta
- (1) GnRH
 - (2) FSH
 - (3) CRH
 - (4) LH
167. During cleavage the cell division is very rapid. The daughter cells do not undergo any growth and the cell thus become gradually smaller in volume Hence :-
- (1) There is no growth in the volume and size of the embryo
 - (2) The embryo grows in volume and same in size
 - (3) The embryo becomes smaller in volume
 - (4) The embryo remains static

168. Read the following statement

- After implantation, finger like projections appear on the trophoblast called chorionic villi
- The blastomeres in the blastocyst are arranged into an outer layer called trophoblast.
- Placenta also acts as an endocrine tissue.
- hCG, hPL and Relaxin are produced in women during pregnancy also.

How many of the above statement are wrong ?

- One
- Three
- Two
- Four

169. Which statement is wrong about FSH. ?

- It induces mitotic division in granulosa cells
- It induces stroma to convert into theca.
- It induces theca interna to secrete estrogen
- It controls progesterone secretion by corpus luteum

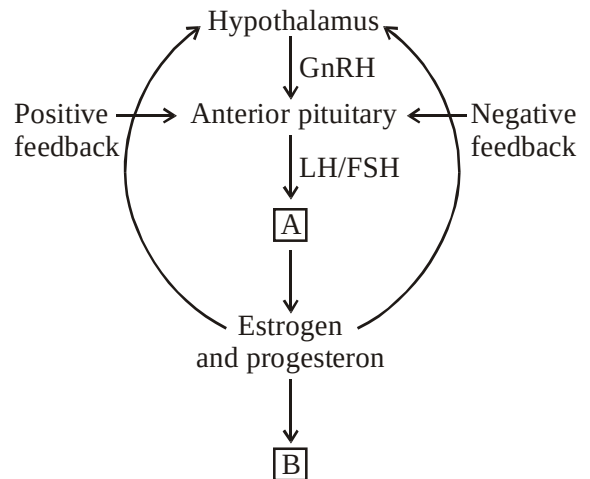
170. Read the following statement :-

- After ovulation a corpus leuteum form from the ruptured follicle and begins to secrete progesterone and estrogen.
- The cavity of cervix is called cervical canal in which mucosol fold present which secrete mucosol secretion.
- The first menses is called menopause and the parmanent cessation of menses is called menarch.
- Nitric oxide cause smooth muscle within erectile tissue to contract which result in widening of blood sinuses of penis.
- Myometrium the layer of uterine linning that is partially shed during menstrual cycle.

How many the statement is/are wrong ?

- four
- Three
- Two
- One

171.



Find out 'A' and 'B' respectively :-

- Uterus, Ovary
- Ovary, Uterus
- Oviduct, ovary
- Seminal vesicle, uterus

172. Arrange these reproductive events occuring in human from last to first and find out the correct sequence :-

Insemination (A), Parturition (B), Formation of blastocyst (C), Gametogenesis (D), Gestation (E), Fertilisation (F)

- D, A, F, C, E, B
- D, F, A, B, C, E
- B, E, C, F, A, D
- B, C, E, F, A, D

173. Menstruation occurs due to decline of :-

- Estrogen
- LH
- FSH
- Progestrone

174. Corpus luteum secretes large amount of progesterone which is essential for :-

- Contraction of the uterus
- Development of a new follicle
- Maintenance of the endometrium
- All of these

175. The proliferative phase in the human menstrual cycle is also called :-

- (1) Follicular phase and lasts for about 9 days
- (2) Secretory phase and lasts for about 9 days
- (3) Follicular phase and lasts for about 13 days
- (4) Luteal phase and lasts for about 6 days

176. Graafian follicle is :-

- (1) Fully developed tertiary follicle
- (2) Mature follicle containing secondary oocyte
- (3) Secondary follicle after development
- (4) All of these

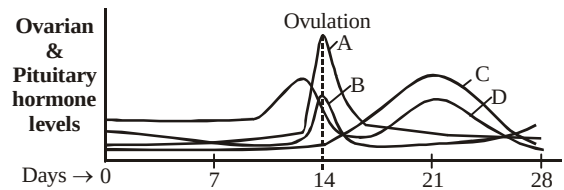
177. Luteal phase is also known as :-

- (1) Follicular phase and characterised by corpus luteum
- (2) Proliferative phase and characterised by graafian follicle
- (3) Secretory phase and characterised by peak of progesterone
- (4) Secretory phase and characterised by regressing graafian follicle

178. How many eggs are released by a human ovary in a year and how many eggs do you think would have been released if the mother gave birth to 3 identical offsprings ?

- | | |
|-----------|-----------|
| (1) 12, 3 | (2) 6, 3 |
| (3) 6, 1 | (4) 12, 1 |

179. In given graph A to D represents respectively



- (1) FSH, LH, Estrogen, Progesterone
- (2) LH, Estrogen, FSH, Progesterone
- (3) LH, FSH, Progesterone, Estrogen
- (4) LH, Progesterone, FSH, Estrogen

180. Which of the following statement is incorrect ?

- (1) Menstrual cycle starts only after attaining sexual maturation
- (2) During ovulation only one ovum is released per menstrual cycle
- (3) The presence of X or Y chromosome in the ovum determines the sex of the embryo
- (4) Inner cell mass of blastocyst contains certain cells called stem cells which have the potency to give rise to all the tissues and organs

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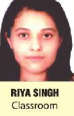
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